

**Ex.No:9****Date:30/09/25**

## **DESIGN A SIMPLE TOPOLOGY USING CISCO PACKET TRACER**

### **Aim:**

To design and configure a simple network topology with one router, two switches, and four PCs in Cisco Packet Tracer for successful inter-network communication.

### **Procedure:**

#### **1. Launch Cisco Packet Tracer**

- Open Cisco Packet Tracer on your system.
- Ensure the workspace and device categories (Routers, Switches, End Devices, etc.) are visible.

#### **2. Add Network Devices**

- From the Routers section, drag one router onto the workspace.
- From the Switches section, drag two 2960 switches.
- From the End Devices section, drag four PCs (PC1, PC2, PC3, PC4).
- Rename the devices.



### 3. Connect the Devices

Use Copper Straight-Through cables to connect:

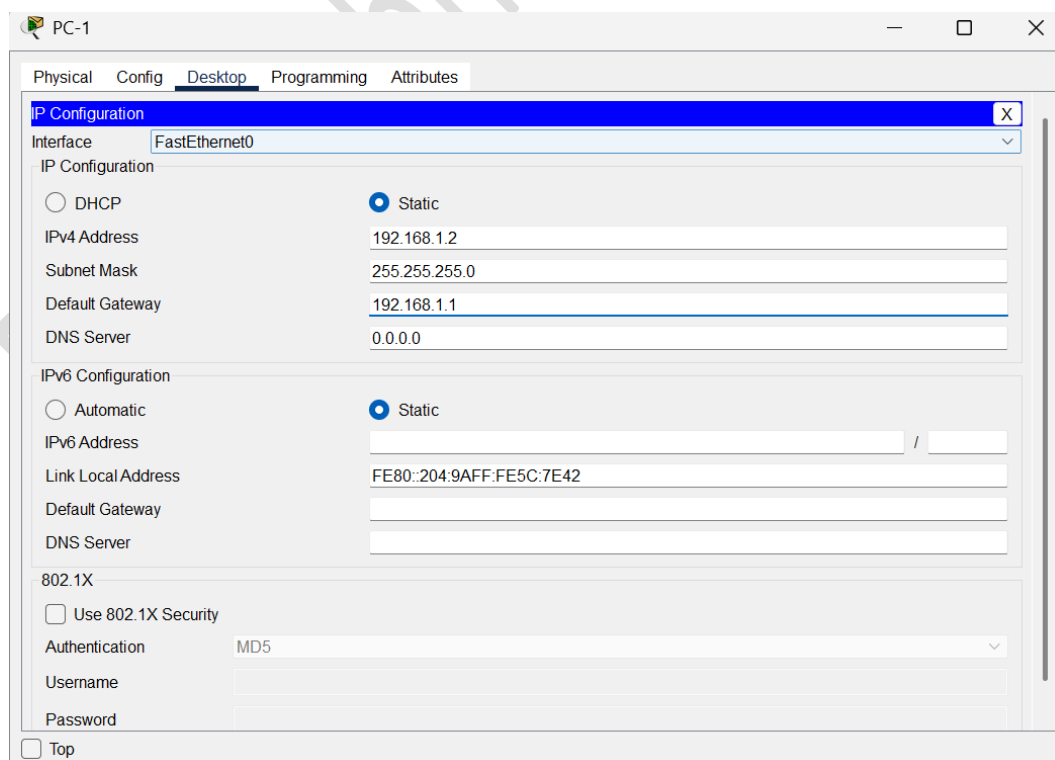
- PC1 to Switch1 using port FastEthernet0/1.
- PC2 to Switch1 using port FastEthernet0/2.
- PC3 to Switch2 using port FastEthernet0/1.
- PC4 to Switch2 using port FastEthernet0/2.
- Switch1 to the router using Switch1 port GigabitEthernet0/1 and Router port GigabitEthernet0/0/0.
- Switch2 to the router using Switch2 port GigabitEthernet0/1 and Router port GigabitEthernet0/0/1.

### 4. Assign IP Addresses:

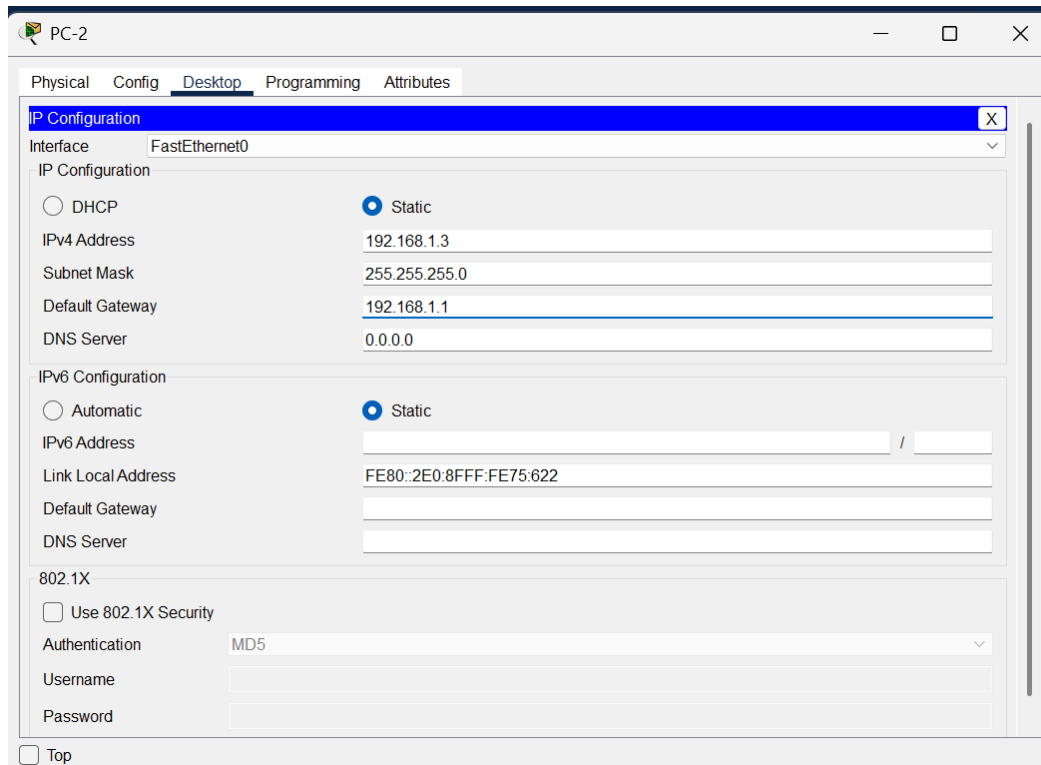
- Network 1 (Switch1 side): 192.168.1.0/24
- Network 2 (Switch2 side): 192.168.2.0/24

**For each PC (PC1 to PC4):**

1. Click the PC → go to Desktop → IP Configuration.
  2. Select Static and manually enter the IP address, subnet mask, and default gateway
- PC1 → IP 192.168.1.2 / 255.255.255.0, Gateway 192.168.1.1



- PC2 → IP 192.168.1.3 / 255.255.255.0, Gateway 192.168.1.1

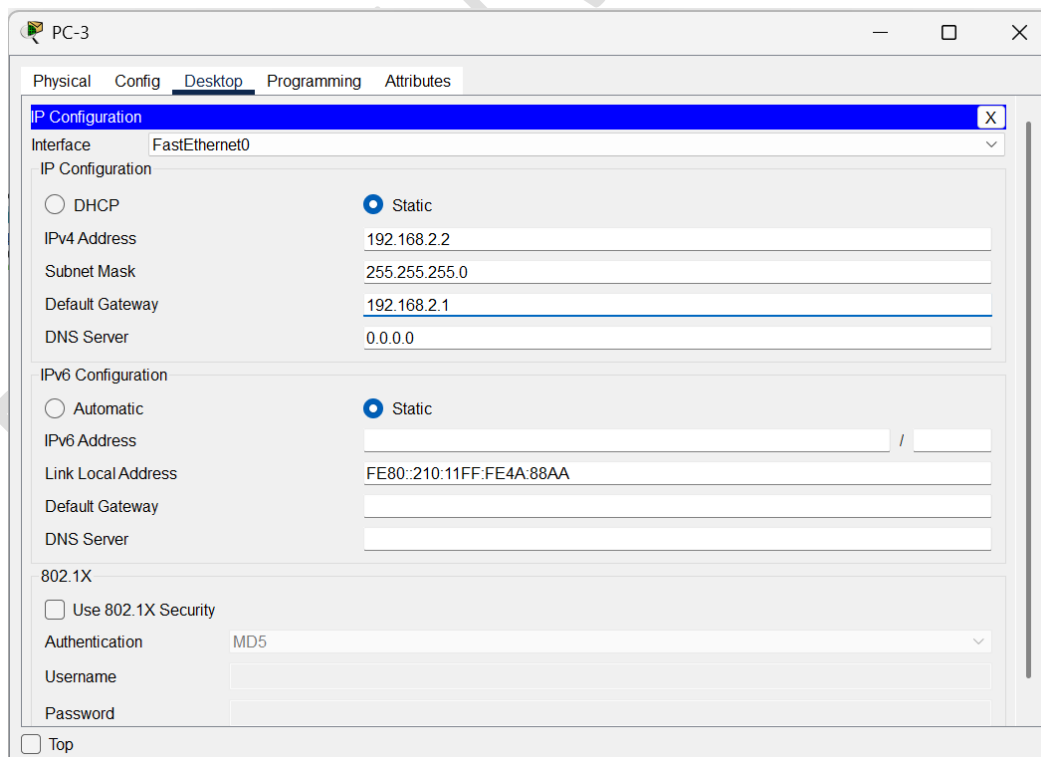


The screenshot shows the 'PC-2' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing the 'Interface' as 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IPv4 Address: 192.168.1.3, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1, and DNS Server: 0.0.0.0. The 'IPv6 Configuration' section shows 'Static' selected with a Link Local Address of FE80::2E0:8FFF:FE75:622. The '802.1X' section has 'Use 802.1X Security' unchecked, 'Authentication' set to MD5, and empty fields for Username and Password. A 'Top' button is at the bottom left.

| IP Configuration                             |                                         |
|----------------------------------------------|-----------------------------------------|
| Interface: FastEthernet0                     |                                         |
| IP Configuration                             |                                         |
| <input type="radio"/> DHCP                   | <input checked="" type="radio"/> Static |
| IPv4 Address                                 | 192.168.1.3                             |
| Subnet Mask                                  | 255.255.255.0                           |
| Default Gateway                              | 192.168.1.1                             |
| DNS Server                                   | 0.0.0.0                                 |
| IPv6 Configuration                           |                                         |
| <input type="radio"/> Automatic              | <input checked="" type="radio"/> Static |
| IPv6 Address                                 |                                         |
| Link Local Address                           | FE80::2E0:8FFF:FE75:622                 |
| Default Gateway                              |                                         |
| DNS Server                                   |                                         |
| 802.1X                                       |                                         |
| <input type="checkbox"/> Use 802.1X Security |                                         |
| Authentication                               | MD5                                     |
| Username                                     |                                         |
| Password                                     |                                         |

☐ Top

- PC3 → IP 192.168.2.2 / 255.255.255.0, Gateway 192.168.2.1

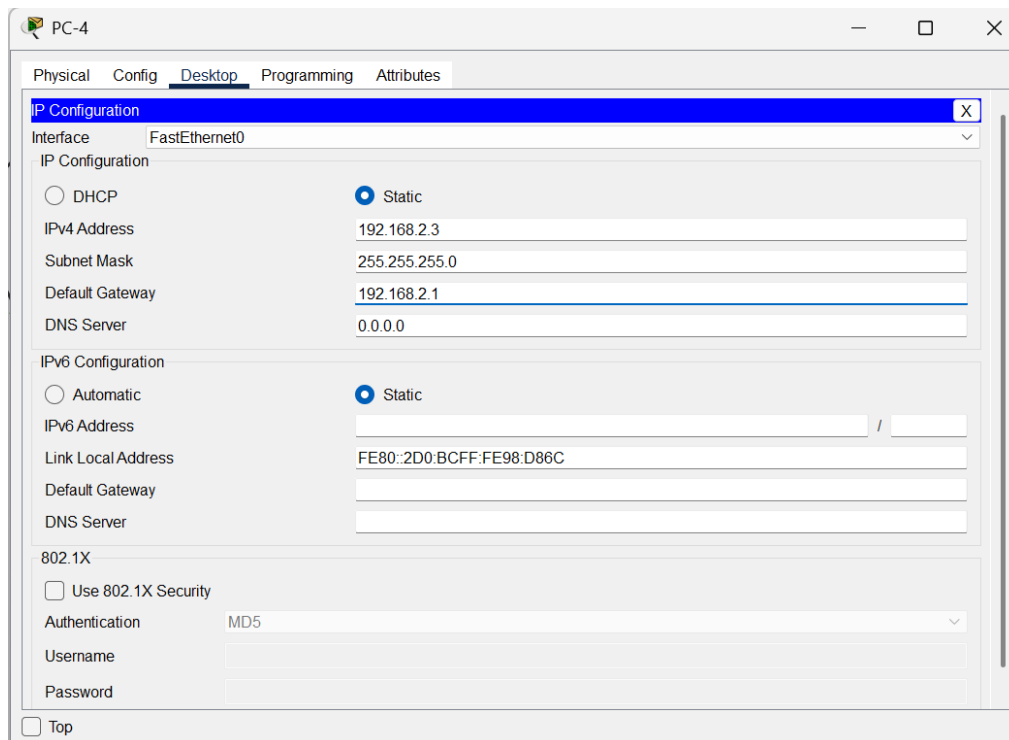


The screenshot shows the 'PC-3' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing the 'Interface' as 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IPv4 Address: 192.168.2.2, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.2.1, and DNS Server: 0.0.0.0. The 'IPv6 Configuration' section shows 'Static' selected with a Link Local Address of FE80::210:11FF:FE4A:88AA. The '802.1X' section has 'Use 802.1X Security' unchecked, 'Authentication' set to MD5, and empty fields for Username and Password. A 'Top' button is at the bottom left.

| IP Configuration                             |                                         |
|----------------------------------------------|-----------------------------------------|
| Interface: FastEthernet0                     |                                         |
| IP Configuration                             |                                         |
| <input type="radio"/> DHCP                   | <input checked="" type="radio"/> Static |
| IPv4 Address                                 | 192.168.2.2                             |
| Subnet Mask                                  | 255.255.255.0                           |
| Default Gateway                              | 192.168.2.1                             |
| DNS Server                                   | 0.0.0.0                                 |
| IPv6 Configuration                           |                                         |
| <input type="radio"/> Automatic              | <input checked="" type="radio"/> Static |
| IPv6 Address                                 |                                         |
| Link Local Address                           | FE80::210:11FF:FE4A:88AA                |
| Default Gateway                              |                                         |
| DNS Server                                   |                                         |
| 802.1X                                       |                                         |
| <input type="checkbox"/> Use 802.1X Security |                                         |
| Authentication                               | MD5                                     |
| Username                                     |                                         |
| Password                                     |                                         |

☐ Top

- PC4 → IP 192.168.2.3 / 255.255.255.0, Gateway 192.168.2.1



### For Router:

- Router interface GigabitEthernet0/0/0 → 192.168.1.1 / 255.255.255.0
- Router interface GigabitEthernet0/0/1 → 192.168.2.1 / 255.255.255.0

Click on Router, go to the CLI tab, and enter the following commands:

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

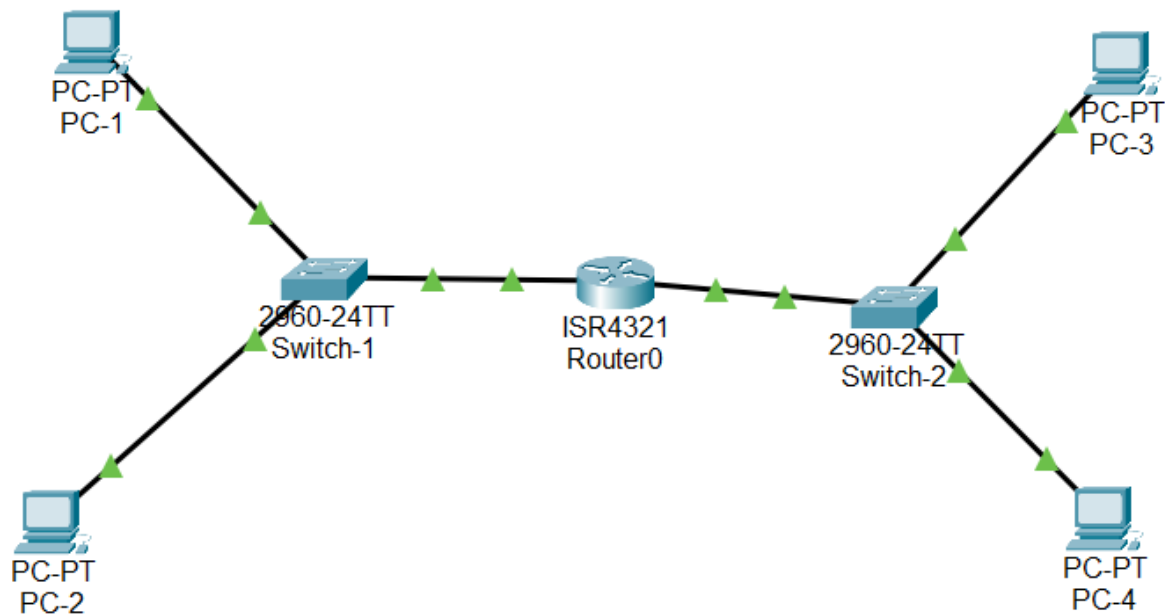
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#show ip interface brief

```

| Interface            | IP-Address  | OK? | Method | Status                | Protocol |
|----------------------|-------------|-----|--------|-----------------------|----------|
| GigabitEthernet0/0/0 | 192.168.1.1 | YES | manual | up                    | up       |
| GigabitEthernet0/0/1 | 192.168.2.1 | YES | manual | up                    | up       |
| Vlan1                | unassigned  | YES | unset  | administratively down | down     |

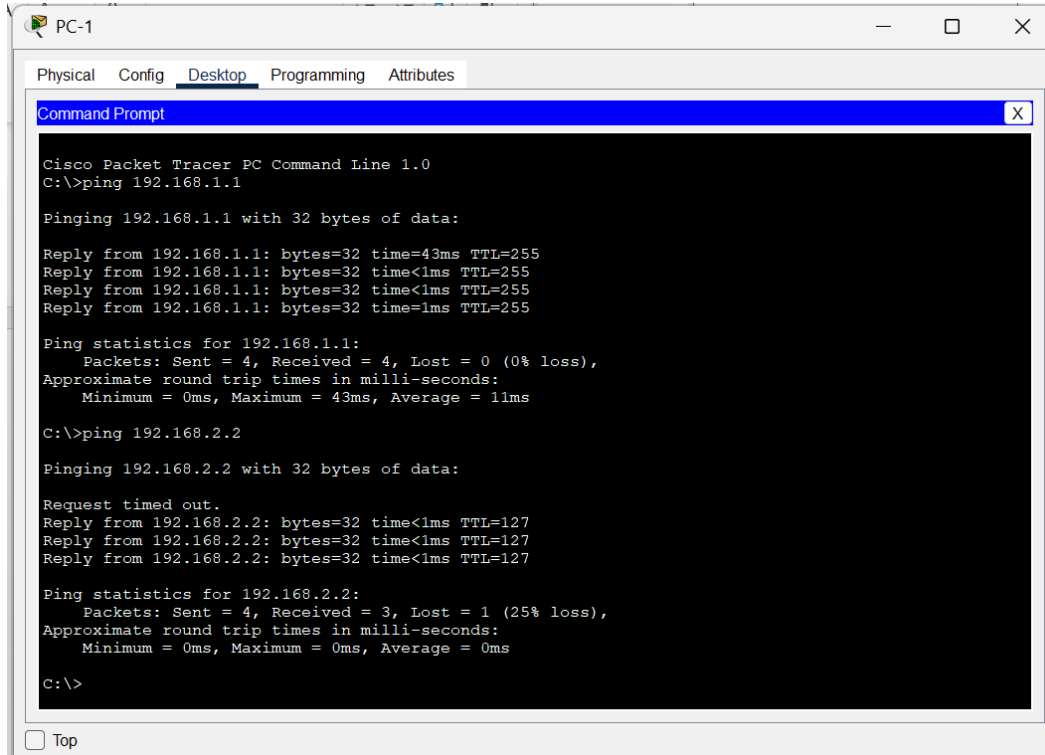
```
Router#
```



## 5. Test Connectivity

These test the router and cross-network connectivity.

- On PC1, open the Command Prompt and type:



The screenshot shows the Command Prompt window for PC-1 in Cisco Packet Tracer. The window title is 'PC-1' and it has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, showing a 'Command Prompt' window. The command prompt displays the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=43ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 43ms, Average = 11ms

C:\>ping 192.168.2.2

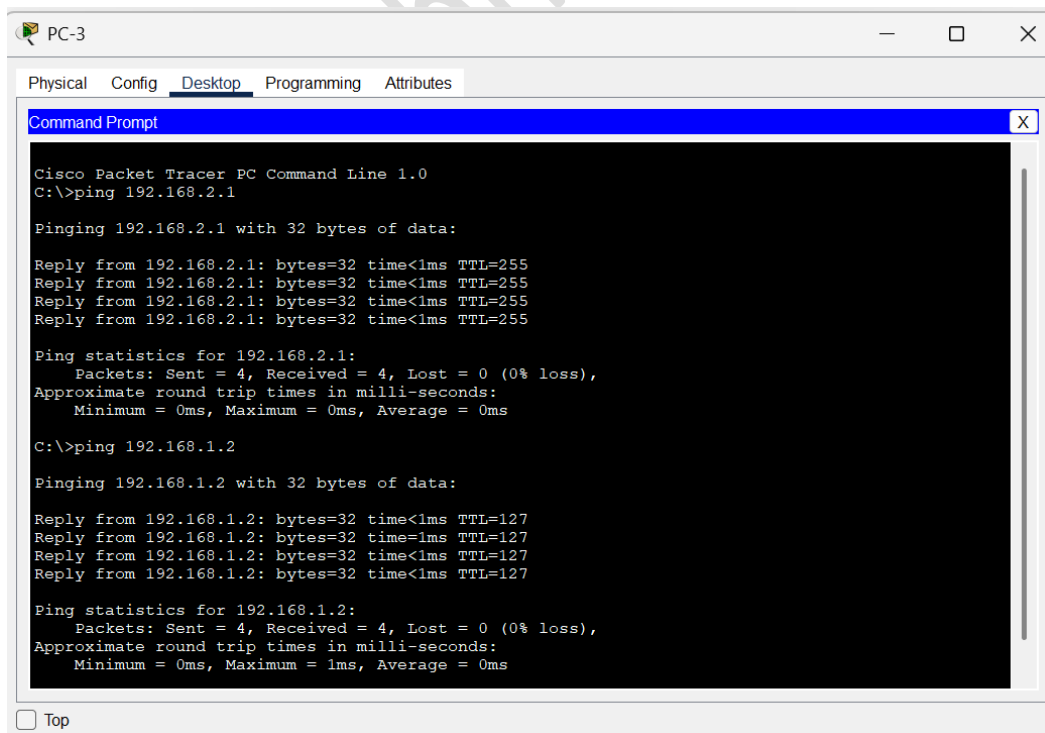
Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

- On PC3, open the Command Prompt and type:



The screenshot shows the Command Prompt window for PC-3 in Cisco Packet Tracer. The window title is 'PC-3' and it has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, showing a 'Command Prompt' window. The command prompt displays the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time=1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Successful replies confirm that inter-network communication is functioning correctly.

### **Result:**

Thus, a simple network topology with one router, two switches, and four PCs was successfully designed and configured in Cisco Packet Tracer. Successful ping replies verified proper connectivity between all devices in both networks.